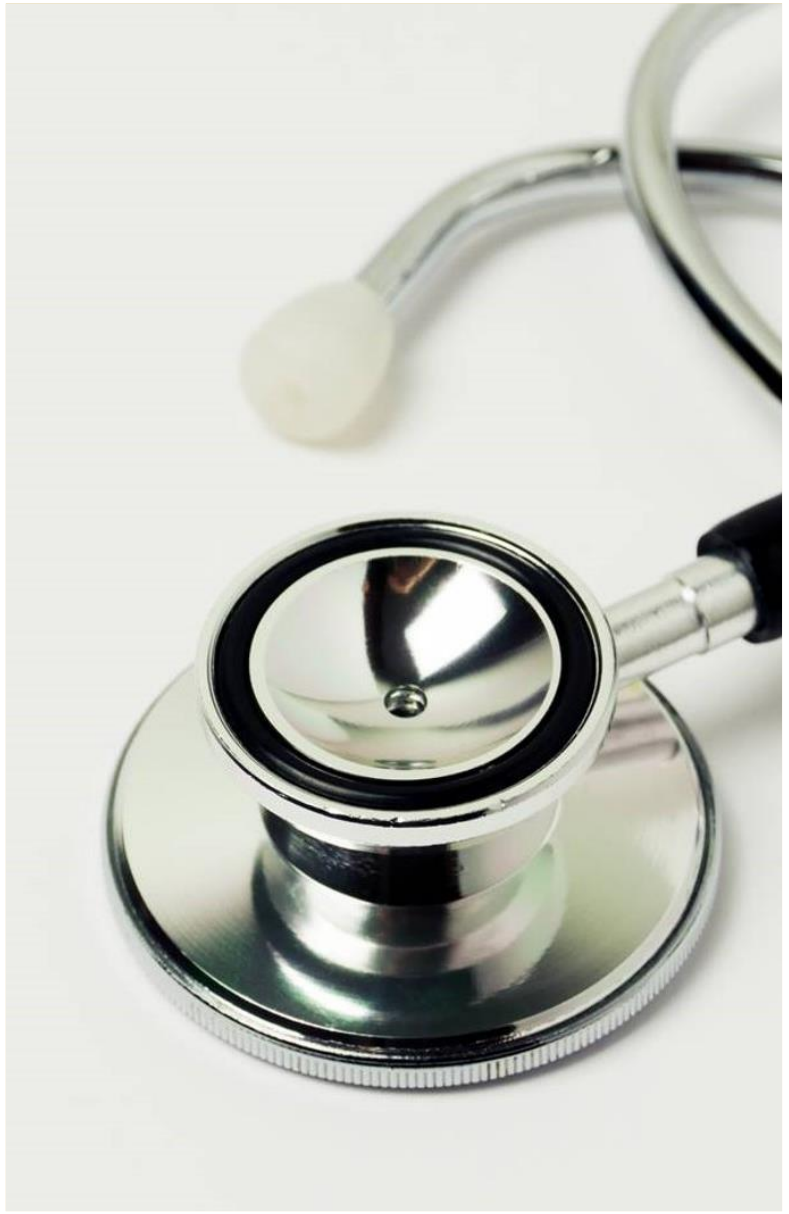
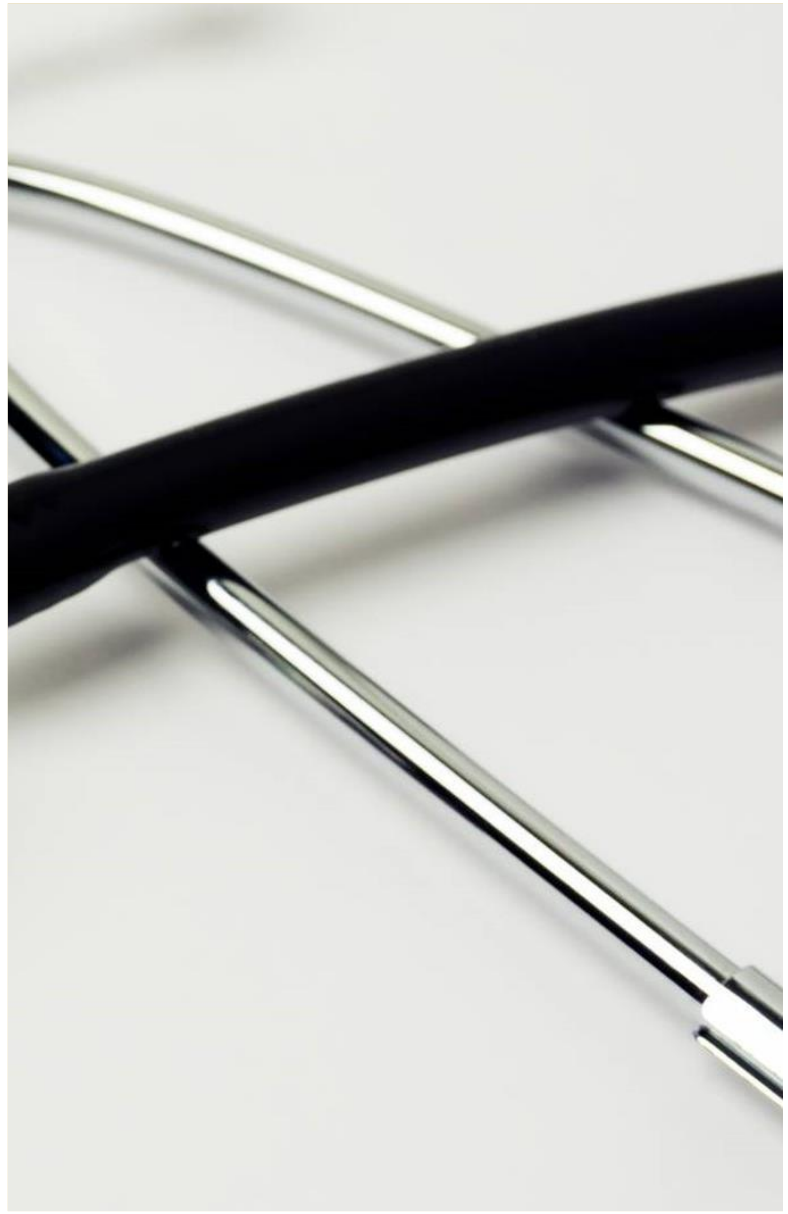
WHITE BLOOD DISORDERS



AGENDA

1. Lymphoma
2. Leukemia
3. Multiple myeloma





INTRODUCTION

- Types of WBCs: Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils
- Functions: Immunity, infection defense, inflammation response
- Normal WBC count: 4,000-11,000 cells/µL

WHITE BLOOD CELL DISORDERS OVERVIEW

- **Neoplastic Disorders:** Uncontrolled WBC proliferation (e.g., leukemias, lymphomas)
- **Non-neoplastic Disorders:** WBC dysfunction due to infections, medications, or immune deficiencies
- **Symptoms:** Fatigue, infections, swollen lymph nodes, night sweats, weight loss

Hodgkin lymphoma

- **Definition:** Cancer of lymphoid tissue characterized by Reed-Sternberg cells
- **Risk Factors:** EBV infection, immunosuppression, family history
- **Symptoms:** Painless swollen lymph nodes, fever, night sweats, weight loss

DIAGNOSIS & TREATMENT OF HODGKIN LYMPHOMA

- **Diagnosis:** Lymph node biopsy, CT scan, PET scan
- **Staging:** Ann Arbor staging system
- **Treatment:** Chemotherapy (ABVD regimen), radiation, immunotherapy

NON-HODGKIN LYMPHOMA (NHL)

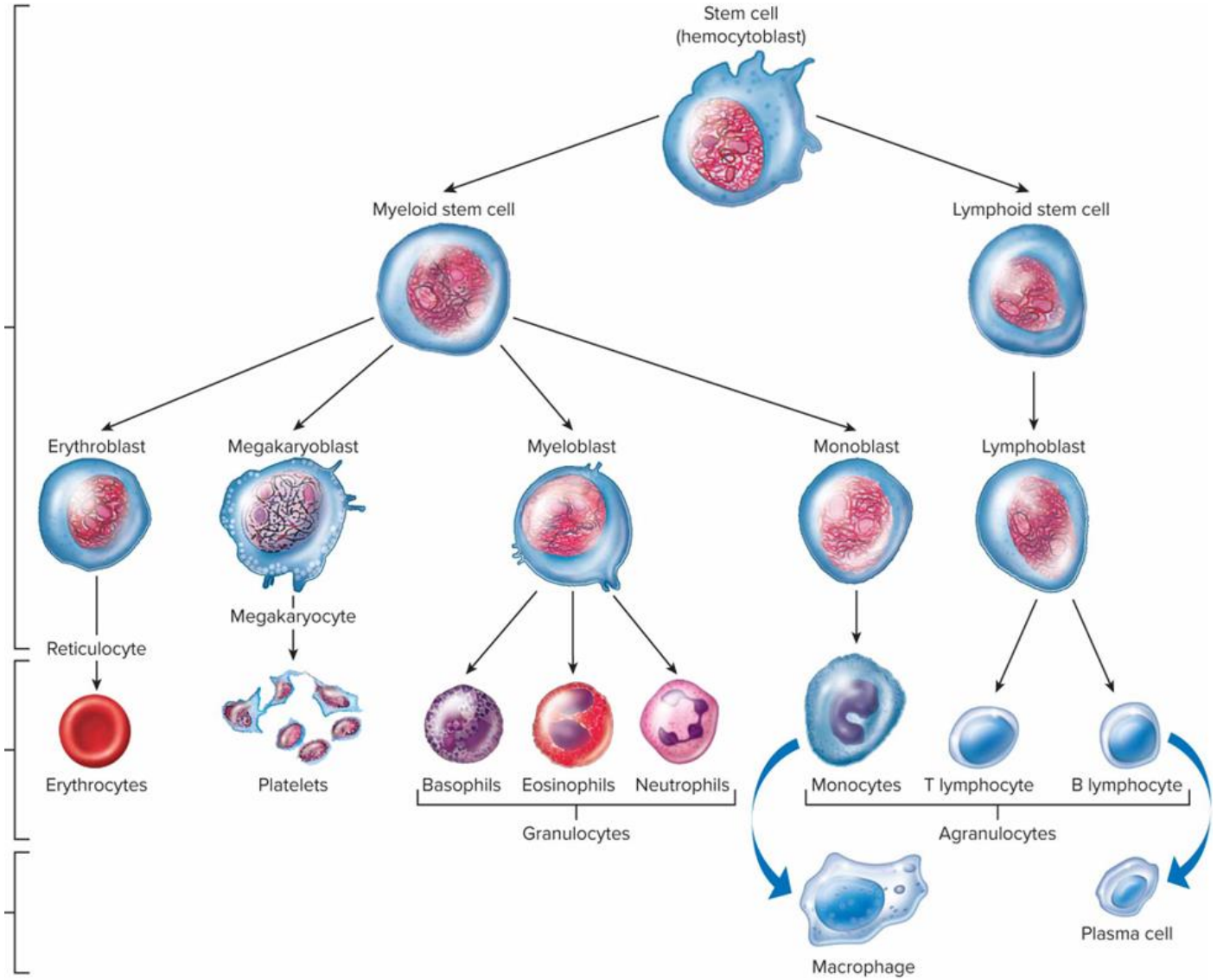
- **Definition:** A diverse group of lymphoid cancers without Reed-Sternberg cells
- **Types:** B-cell NHL (e.g., Diffuse Large B-cell Lymphoma), T-cell NHL
- **Risk Factors:** Immunosuppression, EBV, autoimmune diseases
- **Symptoms:** Swollen lymph nodes, fever, night sweats, weight loss

DIAGNOSIS & TREATMENT OF NHL

- **Diagnosis:** Lymph node biopsy, immunophenotyping (CD markers), imaging
- **Treatment:** Chemotherapy (R-CHOP regimen for B-cell NHL), radiation, targeted therapy

LEUKEMIAS – OVERVIEW

- **Definition:** Uncontrolled WBC proliferation in bone marrow
- **Types:**
  - Acute vs. Chronic
  - Lymphoid vs. Myeloid
- **Symptoms:** Fatigue, frequent infections, easy bruising/bleeding


Stem cell
(hemocytoblast)
Myeloid stem cell
Lymphoid stem cell
Erythroblast
Megakaryoblast
Myeloblast
Monoblast
Lymphoblast
Reticulocyte
Megakaryocyte
Erythrocytes
Platelets
Basophils
Eosinophils
Neutrophils
Granulocytes
Monocytes
T lymphocyte
B lymphocyte
Agranulocytes
Macrophage
Plasma cell

ACUTE LEUKEMIAS (ALL & AML)

- **ALL (Acute Lymphoblastic Leukemia):** Most common in children, aggressive
- **AML (Acute Myeloid Leukemia):** More common in adults, associated with Auer rods

CHRONIC LEUKEMIAS (CLL & CML)

- **CLL (Chronic Lymphocytic Leukemia):** Slow-growing, seen in older adults
- **CML (Chronic Myeloid Leukemia):** Linked to the Philadelphia chromosome

DIAGNOSIS & TREATMENT OF LEUKEMIAS

- **Diagnosis:** Blood smear, bone marrow biopsy, genetic testing
- **Treatment:**
  - Chemotherapy
  - Targeted therapy (e.g., Imatinib for CML)
  - Bone marrow transplantation

MULTIPLE MYELOMA

- **Definition:** Cancer of plasma cells in bone marrow
- **Key Feature:** Excessive production of **abnormal antibodies (M protein)**
- **Symptoms (CRAB Criteria):**
  - **C**alcium elevation (hypercalcemia)
  - **R**enal dysfunction
  - **A**nemia
  - **B**one pain & fractures

DIAGNOSIS & TREATMENT OF MULTIPLE MYELOMA

- **Diagnosis:** Serum protein electrophoresis (M spike), bone marrow biopsy
- **Treatment:** Chemotherapy, stem cell transplant, immunomodulatory drugs

LABORATORY TESTS FOR WBC DISORDERS

- Complete Blood Count (CBC)
- Peripheral Blood Smear
- Bone Marrow Aspiration & Biopsy
- Flow Cytometry & Molecular Markers

THANK YOU

Yasir Eltahir

